Lecture 15

The Chain Rule

Chain Rules for Derivatives:

For single variable functions

For a function $y = f(x)$, where y is a differentiable function of x and x itself is a differentiable function of the parameter t, then, under composition, y becomes a differentiable function of t with,

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

For multivariate functions

If $x = x(t)$ and $y = y(t)$ are differentiable at t, and if $z = f(x, y)$ is differentiable at the point $(x, y) = \big(x(t), y(t)\big)$, then $z = f\big(x(t), y(t)\big)$ is differentiable at t and

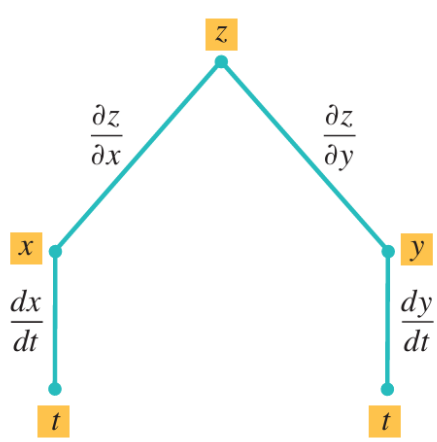
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

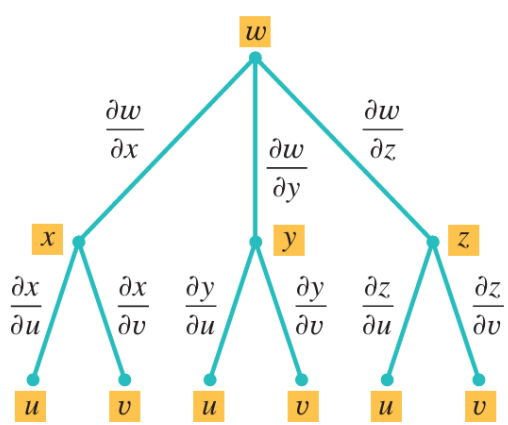
where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y).

If each of the functions $x = x(t)$, $y = y(t)$ and $z = z(t)$ is differentiable at t and $w = f(x, y, z)$ is differentiable at the point $(x, y, z) = \big(x(t), y(t), z(t)\big)$, then the function $w = f\big(x(t), y(t), z(t)\big)$ is differentiable at t and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y, z).





Example: If $f(x) = 3x^2 + 2$, where $x = 3\cos\theta$, find $\frac{df}{d\theta}$.

Solution: We know, here,

$$\frac{df}{d\theta} = \frac{df}{dx} \cdot \frac{dx}{d\theta}$$

$$= 6x \cdot (-3\sin\theta)$$

$$= -18x\sin\theta$$

$$= -54\cos\theta\sin\theta$$

Example: If $z = f(x, y) = x^2e^y$, $x(t) = t^2 - 1$, $y(t) = \sin t$, find $\frac{dz}{dt}$.

Solution: We know, here,

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$= 2xe^y \cdot 2t + x^2e^y \cdot \cos t$$

$$= 2(t^2 - 1)e^{\sin t} \cdot 2t + (t^2 - 1)^2e^{\sin t}\cos t$$

$$= 4t(t^2 - 1)e^{\sin t} + (t^2 - 1)^2e^{\sin t}\cos t$$

Example: Suppose $w = \sqrt{x^2 + y^2 + z^2}$, $x = \cos\theta$, $y = \sin\theta$, $z = \tan\theta$. Use chain rule to find $\frac{dw}{d\theta}$ at $\theta = \frac{\pi}{4}$.

Solution: By chain rule, we know that,

$$\frac{dw}{d\theta} = \frac{\partial w}{\partial x} \cdot \frac{dx}{d\theta} + \frac{\partial w}{\partial y} \cdot \frac{dy}{d\theta} + \frac{\partial w}{\partial z} \cdot \frac{dz}{d\theta}$$

$$= \frac{1}{2}(x^2 + y^2 + z^2)^{\frac{1}{2}-1}(2x)(-\sin\theta) + \frac{1}{2}(x^2 + y^2 + z^2)^{\frac{1}{2}-1}(2y)(\cos\theta) + \frac{1}{2}(x^2 + y^2 + z^2)^{\frac{1}{2}-1}(2z)(\sec^2\theta)$$

$$= (x^2 + y^2 + z^2)^{-\frac{1}{2}}x(-\sin\theta) + (x^2 + y^2 + z^2)^{-\frac{1}{2}}y(\cos\theta) + (x^2 + y^2 + z^2)^{-\frac{1}{2}}z(\sec^2\theta)$$

$$= \frac{-x\sin\theta}{\sqrt{x^2+y^2+z^2}} + \frac{y\cos\theta}{\sqrt{x^2+y^2+z^2}} + \frac{z\sec^2\theta}{\sqrt{x^2+y^2+z^2}}$$

$$= \frac{1}{\sqrt{x^2 + y^2 + z^2}}(-x\sin\theta + y\cos\theta + z\sec^2\theta).$$

At $\theta = \frac{\pi}{4}$, we have

$$x = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}, \qquad y = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}, \qquad z = \tan\frac{\pi}{4} = 1.$$

Therefore,

$$\left.\frac{dw}{d\theta}\right|_{\theta=\pi/4} = \frac{1}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + 1^2}}\left(-\frac{1}{\sqrt{2}}\sin\frac{\pi}{4} + \frac{1}{\sqrt{2}}\cos\frac{\pi}{4} + 1\cdot\left(\sec\frac{\pi}{4}\right)^2\right)$$

$$= \frac{1}{\sqrt{\frac{1}{2}+\frac{1}{2}+1}}\left(-\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}} + 1\cdot 2\right)$$

$$= \frac{1}{\sqrt{2}}\left(-\frac{1}{2}+\frac{1}{2}+2\right)$$

$$= \frac{2}{\sqrt{2}}$$

$$= \sqrt{2} \quad \text{(Ans.)}$$

H.W. Given $z = 3x^3\sin y + 5y^3\cos x$, where $x = 2\cos t$ and $y = 2\sin t$, find $\frac{dz}{dt}$.

Answer: $-144\sin t\cdot\cos^2 t\cdot\sin(2\sin t) + 80\sin^4 t\cdot\sin(2\cos t) + 48\cos^4 t\cdot\cos(2\sin t) + 120\sin^2 t\cos t\cdot\cos(2\cos t)$.

Chain Rules for Derivatives:

- If $x = x(u,v)$ and $y = y(u,v)$ have first-order partial derivatives at the point (u,v) and if $z = f(x,y)$ is differentiable at the point $(x,y) = \big(x(u,v),\, y(u,v)\big)$, then $z = f\big(x(u,v),\, y(u,v)\big)$ has first order partial derivatives at the point (u,v), given by,

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} \quad \text{and}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

- If each function $x = x(u,v)$, $y = y(u,v)$ and $z = z(u,v)$ has first order partial derivatives at the point (u,v) and if the function $w = f(x,y,z)$ is differentiable at the point $(x,y,z) = \big(x(u,v),\, y(u,v),\, z(u,v)\big)$, then $w = f\big(x(u,v),\, y(u,v),\, z(u,v)\big)$ has first order partial derivatives at the point (u,v) given by,

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial u} \quad \text{and}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial v}$$

Example: Suppose that $z = f(x, y)$, $f(x, y) = e^{xy}$, $x(u, v) = 3u\sin v$ and $y(u, v) = 4uv^2$. Then find the partial derivatives $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.

Solution: Given,

$z = f(x, y) = e^{xy}$

$x(u, v) = 3u\sin v$

$y(u, v) = 4uv^2$

Differentiating z with respect to u, we get,

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$

$$= e^{xy}.y.\ (3\sin v) + e^{xy}.\ x.\ (4v^2)$$

$$= 3ye^{xy}\sin v + 4xe^{xy}v^2$$

$$= e^{xy}\ [3y\sin v + 4xv^2]$$

$$= e^{(3u\sin v)(4uv^2)}[3(4uv^2)\sin v + 4(3u\sin v)v^2]$$

$$= e^{12u^2v^2\sin v}\ (12uv^2\sin v + 12uv^2\sin v)$$

$$= e^{12u^2v^2\sin v}\ 24uv^2\sin v$$

$$= 24uv^2\sin v\, e^{12u^2v^2\sin v}$$

Differentiating z with respect to v,

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

$$= e^{xy}.\ y.(3u\cos v) + e^{xy}.\ x.(4u \cdot 2v)$$

$$= e^{xy}\ (3yu\cos v + 8xuv)$$

$$= e^{(3u\sin v)(4uv^2)}(3(4uv^2)u\cos v + 8(3u\sin v)uv)$$

$$= e^{12u^2v^2\sin v}(12u^2v^2\cos v + 24u^2v\sin v)$$

Example: For a differentiable function $f(x, y)$ with continuous second partial derivatives where $x = r\cos\theta, y = r\sin\theta$, then show that,

i. $f_r = f_x\cos\theta + f_y\sin\theta$
ii. $f_\theta = -rf_x\sin\theta + rf_y\cos\theta$
iii. $f_{rr} = f_{xx}\cos^2\theta + 2f_{xy}\cos\theta\sin\theta + f_{yy}\sin^2\theta$
iv. $f_{\theta\theta} = r^2f_{xx}\sin^2\theta - 2r^2f_{xy}\cos\theta\sin\theta + r^2f_{yy}\cos^2\theta - rf_x\cos\theta - rf_y\sin\theta.$

Solution:

i. $f_r = \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r}$
$= f_x\cos\theta + f_y\sin\theta$ [Showed]

ii. $f_\theta = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}$
$= f_x(-r\sin\theta) + f_y(r\cos\theta)$
$= -rf_x\sin\theta + rf_y\cos\theta$ [Showed]

iii. $f_{rr} = \frac{\partial^2 f}{\partial r^2} = \frac{\partial}{\partial r}\left(\frac{\partial f}{\partial r}\right)$

$$= \frac{\partial}{\partial r}\left(f_x\cos\theta + f_y\sin\theta\right)$$
$$= \frac{\partial(f_x)}{\partial r}\cos\theta + \frac{\partial(f_y)}{\partial r}\sin\theta$$
$$= \left(\frac{\partial f_x}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f_x}{\partial y}\frac{\partial y}{\partial r}\right)\cos\theta + \left(\frac{\partial f_y}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f_y}{\partial y}\frac{\partial y}{\partial r}\right)\sin\theta$$
$$= \left(f_{xx}\cos\theta + f_{xy}\sin\theta\right)\cos\theta + \left(f_{xy}\cos\theta + f_{yy}\sin\theta\right)\sin\theta$$
$$= f_{xx}\cos^2\theta + 2f_{xy}\cos\theta\sin\theta + f_{yy}\sin^2\theta.$$ [Showed]

iv. $f_{\theta\theta} = \frac{\partial^2 f}{\partial \theta^2} = \frac{\partial}{\partial \theta}\left(\frac{\partial f}{\partial \theta}\right)$

$$= \frac{\partial}{\partial \theta}\left(-rf_x\sin\theta + rf_y\cos\theta\right)$$
$$= \frac{\partial(f_x)}{\partial \theta}(-r\sin\theta) + (-rf_x)\frac{\partial}{\partial \theta}(\sin\theta) + \frac{\partial(f_y)}{\partial \theta}(r\cos\theta) + \left(rf_y\right)\frac{\partial}{\partial \theta}(\cos\theta)$$
$$= \left[\frac{\partial f_x}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f_x}{\partial y}\frac{\partial y}{\partial \theta}\right](-r\sin\theta) - rf_x\cos\theta + \left[\frac{\partial f_y}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f_y}{\partial y}\frac{\partial y}{\partial \theta}\right](r\cos\theta) - rf_y\sin\theta$$
$$= \left[f_{xx}(-r\sin\theta) + f_{xy}(r\cos\theta)\right](-r\sin\theta) - rf_x\cos\theta + \left[f_{xy}(-r\sin\theta) + f_{yy}(r\cos\theta)\right](r\cos\theta) - rf_y\sin\theta$$
$$= r^2f_{xx}\sin^2\theta - 2r^2f_{xy}\cos\theta\sin\theta + r^2f_{yy}\cos^2\theta - rf_x\cos\theta - rf_y\sin\theta.$$ [Showed]

Example: Let $f(x,y,z) = \ln(x+y), x = u^2v, y = uv^3$. Find $\frac{\partial f}{\partial u}$ and $\frac{\partial f}{\partial v}$ at the point $(1,-1)$.

Solution: Differentiating f with respect to u, we get,

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$$
$$= \frac{1}{x+y}(2uv) + \frac{1}{x+y}(v^3)$$
$$= \frac{2uv + v^3}{x+y} = \frac{2uv + v^3}{u^2v + uv^3}$$

At $(u, v) = (1, -1)$,

$$\left.\frac{\partial f}{\partial u}\right|_{u=1,v=-1} = \frac{2(1)(-1) + (-1)^3}{(1)^2(-1) + (1)(-1)^3} = \frac{-2-1}{-1-1} = \frac{-3}{-2} = \frac{3}{2}.$$

Again, Differentiating f with respect to v, we get,

$$\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v}$$

$$= \frac{1}{x+y}(u^2) + \frac{1}{x+y}(3uv^2)$$

$$= \frac{u^2 + 3uv^2}{x+y} = \frac{u^2 + 3uv^2}{u^2v + uv^3}.$$

At $(u, v) = (1, -1)$:

$$\left.\frac{\partial f}{\partial v}\right|_{u=1,v=-1} = \frac{(1)^2 + 3(1)(-1)^2}{(1)^2(-1) + (1)(-1)^3} = \frac{1+3}{-1-1} = \frac{4}{-2} = -2.$$

H.W. Suppose that $w = x^2 + y^2 - z^2$ and $x = \rho\sin\phi\cos\theta, y = \rho\sin\phi\sin\theta, z = \rho\cos\phi$. Find $\frac{\partial w}{\partial \rho}$ and $\frac{\partial w}{\partial \theta}$.

Answer: $\frac{\partial w}{\partial \rho} = -2\rho\cos 2\phi$, $\frac{\partial w}{\partial \theta} = 0$

H.W. 13.5 (1–6), (17–22).